

Emanuel Molla

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EDUCATION

Columbia University, Fu Foundation School for Engineering & Applied Science, New York, NY

Expected May 2029

Major: Biomedical Engineering, Bachelor of Science

Relevant Coursework: Introduction to Computing for Engineers and Applied Scientists (Python)

Annandale High School, Annandale, VA

June 2025

Advanced Diploma | GPA: 4.457

Honors: International Baccalaureate Diploma Recipient, IB Middle Years Programme Certificate Recipient

EXPERIENCE

Carnegie Mellon Computational Biology Pre-College Program, Pittsburgh, PA

June 2024–July 2024

Student

- Conducted microbiome research by isolating and characterizing bacterial colonies from pond samples, performing DNA extraction, PCR amplification of 16S rRNA genes, gel electrophoresis, and Sanger sequencing to determine bacterial taxonomy.
- Applied modern molecular biology techniques, including *E. coli* transfection with a green fluorescent protein gene and automated liquid handling using the Opentrons OT-2 robot, integrating concepts from robotics and machine learning for experimental optimization.
- Collaborated with a group of students to implement fundamental bioinformatics algorithms for microbiome diversity analysis, sequence alignment, genome assembly, and phylogenetic tree generation.

DREAMS Research Lab, Blacksburg, VA

July 2024 - Aug. 2024

Research Assistant

- Developed a Python-based toolpath generator that converts 2D sketches into optimized 3D printable designs for metal additive manufacturing
- Implemented algorithmic centerline extraction and SVG path generation to enable efficient, high-precision tracing by metal 3D printers
- Collaborated with graduate researchers to translate their experimental and design requirements into functional software features, ensuring the software aligned with their unique research objectives

LEADERSHIP

The A-Blast Student Newspaper, Annandale, VA

Sept. 2022–June 2025

Health Editor

- Oversaw the Health section, managing articles, editing submissions, and ensuring factual accuracy on coverage of public health issues, wellness, and health policy impacting the school community.
- Researched and wrote articles on topics related to public health, wellness, and mental health, synthesizing information into accessible content for a broad audience
- Utilized Adobe InDesign to design the layout for the Health page

Columbia University First-Generation Advisory Board, New York, NY

Oct. 2025-Present

First Year Representative

- Represent First-Generation and Low-Income students and voice their concerns in weekly board meetings
- Coordinate with students and relevant offices to develop recommendations, propose new campus initiatives, and provide critical feedback to address disparities in university programs

ACTIVITIES

Columbia University Ethiopian Eritrean Student Association, *Member*, Sept. 2025–Present

Charles Drew Pre-Medical Society, *Member*, Sept. 2025–Present

Lion Health Medical Engineering Club, *Member*, Sept. 2025–Present

SKILLS AND INTERESTS

Computer: Intermediate Adobe InDesign, Proficient Python, Intermediate Golang

Language: Intermediate Spanish